xRFM: An Empirical and Interpretive Study of a Feature-Learning Kernel Machine for Tabular Data

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1. Introduction

Tabular data resists deep learning: targets often jump abruptly along feature axes, scales are heterogeneous, and many columns are uninformative — properties gradient-boosted decision trees (GBDTs) handle naturally [Grinsztajn et al., 2022]. Classical kernel methods do not: the kernel function fixes feature similarity *before* training (so it cannot learn that some features matter more than others), and kernel ridge regression costs $O(n^3)$, becoming intractable above ~70k samples. **xRFM** [Beaglehole et al., 2025] tackles both. An inner Recursive Feature Machine [Radhakrishnan et al., 2024] learns which features matter via the Average Gradient Outer Product (AGOP) — a matrix capturing how sensitive its predictions are to each input — and uses it to reshape the kernel. An outer balanced binary tree splits the data along the top AGOP eigenvector of a locally-fitted RFM, bringing training to $O(n \log n)$ and producing a separate AGOP per leaf for subpopulation interpretability. We test xRFM on five UCI datasets verified absent from both the TALENT benchmark and xRFM's meta-test suite, against XGBoost, Random Forest, and CatBoost, asking three questions: *(i)* does xRFM's accuracy advantage on smooth tabular data hold on independent datasets; *(ii)* is per-leaf AGOP a useful interpretability tool relative to PCA, mutual information, and permutation importance, particularly at distinguishing real features from injected noise; *(iii)* does the $O(n \log n)$ training-time scaling hold empirically? §2 formalises xRFM, §3 presents results, §4 connects findings to theory, §5 concludes.

2. Methodology

2.1 Average Gradient Outer Product (AGOP)

AGOP is the matrix at the heart of feature learning in xRFM. Intuitively, it asks: across the training set, in which input directions does the model's prediction change most? For a trained predictor $\hat{f} : \mathbb{R}^d \to \mathbb{R}$ and training set $\mathcal{S} = \{x^{(1)}, \dots, x^{(n)}\}$, AGOP is computed by taking the gradient of $\hat{f}$ at every training point, then averaging the outer products:

$$\mathrm{AGOP}(\hat{f}, \mathcal{S}) \;=\; \frac{1}{n} \sum_{i=1}^{n} \nabla \hat{f}\big(x^{(i)}\big)\, \nabla \hat{f}\big(x^{(i)}\big)^{\top} \;\in\; \mathbb{R}^{d \times d}. \tag{1}$$

The result is a $d \times d$ matrix M whose entries summarise the model's sensitivity. For any unit vector v, the quadratic form $v^\top M v$ measures how strongly $\hat{f}$ responds when its input moves along direction v. Two readouts of M answer different questions:

- **Diagonal entries** M_{ii} give per-feature sensitivity. A large M_{ii} means the model's output changes sharply when feature i changes, holding the others fixed — a feature-importance score conceptually similar to a variance-based first-order sensitivity index.
- **Top eigenvectors of** M give the directions in feature space that the model actually uses (the *active subspace* of Constantine et al., 2014). The top eigenvector v_1 is a signed linear combination of features — positive and negative loadings push the prediction in opposite directions — capturing *joint* structure that the diagonal alone misses. Beaglehole et al. (2025, Fig. 7C) demonstrate this on Breast Cancer, where the top eigenvector identifies a signed pair of features driving malignancy scores.

How AGOP compares to other importance methods. The closest alternatives all answer slightly different questions. *PCA* finds high-variance directions of x but ignores y, so its directions need not be predictive at all. *Mutual information* $I(X_i; Y)$ is model-free and easy to interpret per-feature, but cannot capture joint structure and requires density estimation. *Permutation importance* shuffles a feature and measures the loss change — black-box and intuitive, but known to behave poorly when features are correlated. AGOP sits at a different point on the trade-off: it is supervised (it depends on $\hat{f}$), joint (off-diagonal entries encode interactions), and computed in a single forward/backward pass. The cost is that AGOP inherits the model's biases: if $\hat{f}$ overfits noise, AGOP will assign that noise apparent importance. We see exactly this in §3.2.

2.2 The xRFM Algorithm

xRFM nests an iterative kernel-ridge RFM inside the leaves of a balanced binary tree whose internal splits use an AGOP-based supervised criterion.

Tree construction (Alg. A.2). Each internal node decides how to split its data. Given a node containing more than $L_{\max}$ points, xRFM trains a small RFM on a subsample and asks: *which direction in feature space does this RFM care about most?* That direction is the top eigenvector v_1 of the local AGOP. Every point in the node is then projected onto v_1 and split at the **median**, ensuring the tree stays balanced. The procedure recurses until every leaf has at most $L_{\max}$ points. Median splitting is what distinguishes xRFM from prior AGOP-based trees and is what guarantees $O(n \log n)$ training: each level halves the data, and the tree depth is $O(\log n)$.

Leaf training (Alg. A.1). At each leaf, an RFM is fitted using the generalised kernel $K_{p,q}(x, x') = \exp(-\|x - x'\|_p^q / L^q)$ (Schoenberg, 1942) for $0 < q \leq p \leq 2$. Two cases matter for our experiments: $q = 2$ recovers the rotation-invariant Laplace kernel (the library's `l2` mode), and $q = p$ gives an axis-aligned product kernel (`l1`). Starting with $M_0 = I$, the RFM iteration alternates two steps for up to $\tau \leq 10$ rounds: (a) reshape inputs through the current AGOP via $x \to M_t^{1/2} x$; (b) solve kernel ridge regression $(K + \lambda I)\alpha_t = y$ on the reshaped inputs; (c) recompute AGOP from the resulting model; (d) normalise $M \leftarrow M/(\epsilon + \max_{ij} |M_{ij}|)$ to keep its scale bounded. We retain the iteration with the lowest *validation* error rather than running to convergence.

Inference. A test point $x^\star$ is routed through the tree (hard or soft median thresholds) to its leaf ℓ, then predicted as $\hat{y} = K(x^\star \odot M_\ell^{1/2}, X_{M_\ell})\alpha_\ell$. A single tree, with one predictor per test point and no ensembling.

Computational complexity. Training is $O(n \log n)$: leaves cap per-leaf cost at $O(L_{\max}^3)$ (constant in n), and median splits visit $O(n)$ points at each of $O(\log n)$ tree levels. Inference is $O(\log n + L_{\max})$ per sample. A naive kernel method costs $O(n^3)$ training and $O(n^2)$ memory, which becomes intractable above roughly 70,000 samples on a 40 GB GPU. xRFM preserves RFM's feature-learning while bypassing its scaling wall.

2.3 Experimental Design

Datasets. Five UCI datasets (Table 1) verified *absent* from both the TALENT benchmark's 300-dataset list and from xRFM's own meta-test suite. Two regression and three classification tasks, n from 165 to 325,834, d from 12 to 174. The selection deliberately spans different stress points for kernel methods: smooth-target regression (Appliances Energy, Crop Mapping), rule-like axis-aligned targets (Seoul Bike), small-n with high d/n (HCC), and high-d class imbalance with heavy missingness (IDA2016). Appliances Energy serves double duty as our interpretability testbed because it ships with two ground-truth random features.

#	Dataset (UCI ID)	n	d	Task	Features	Key property
1	Seoul Bike Sharing (560)	8,760	12	Regression	9 num + 3 cat	Strong weather × temporal interactions

#	Dataset (UCI ID)	n	d	Task	Features	Key property
2	Appliances Energy (374)	19,735	27	Regression	All numeric	**Contains 2 ground-truth noise features (rv1, rv2)**
3	HCC Survival (423)	165	49	Binary	22 num + 27 cat	Small-n, heavy missingness
4	IDA2016 Scania APS (414)	60,000	170	Binary	All numeric	Class imbalance (~1.6% positive)
5	Crop Mapping, Winnipeg (525)	50,000*	174	Multiclass (7)	All numeric	SAR+optical satellite; *stratified from 325,834

Splits. Stratified 60/20/20 train/val/test with `random_state=42` (stratified on target for classification, random for regression). Results are single-seed; variance estimates would require multiple seeds, an acknowledged limitation.

Preprocessing. Median imputation for numerics, mode imputation for categoricals. For kernel methods (xRFM): standard-scaled numerics + one-hot-encoded categoricals. For tree ensembles: numerics plus ordinal-encoded categoricals (no standardisation, preserving split structure). CatBoost receives native categorical indices.

Hyperparameter tuning. Optuna TPE sampler on the validation split, model-specific trial budgets (xRFM 25, XGBoost 40, Random Forest 20, CatBoost 30). xRFM search space follows Table A.1 of the xRFM paper: bandwidth $L \in [1, 200]$ log-uniform, ridge $\lambda \in [10^{-6}, 1]$ log-uniform, exponent $p \in [0.7, 1.4]$, kernel $\in \{K_{p,2}, K_{p,p}\}$, diagonal $\in \{\text{True}, \text{False}\}$. Final models are trained on the train set and evaluated on held-out test.

Metrics. Regression: RMSE (primary) and R^2. Classification: Accuracy and AUC-ROC (macro-OvR for multiclass). Plus wall-clock training time and per-sample inference time.

Compute. All experiments ran on Modal A10G GPU instances with a CUDA 12.4 / Python 3.12 container. We initially included TabPFN-v2 but the official `tabpfn` package requires interactive license acceptance for weight download; in our headless pipeline we replaced it with CatBoost, which is also more comparable to the other GBDT methods.

3. Results

3.1 Main comparison

Tables 2 and 3 report final test-set metrics for all four models on all five datasets.

Table 2. Regression results (lower RMSE is better).

Dataset	Model	RMSE	R^2	Train (s)	Inf. (μs/sample)
Appliances Energy	xRFM	**65.29**	**0.574**	11.5	26.0
Appliances Energy	XGBoost	65.75	0.568	41.7	94.8
Appliances Energy	CatBoost	66.96	0.552	11.1	3.5
Appliances Energy	RF	72.61	0.473	1.7	41.4
Seoul Bike	CatBoost	**225.66**	**0.878**	5.2	1.9
Seoul Bike	XGBoost	232.32	0.870	5.4	37.2
Seoul Bike	xRFM	238.44	0.864	2.1	15.9
Seoul Bike	RF	245.64	0.855	0.5	57.4

Table 3. Classification results (higher Accuracy / AUC-ROC is better).

Dataset	Model	Accuracy	AUC-ROC	Train (s)	Inf. (μs/sample)
HCC Survival	RF	**0.8485**	**0.8500**	0.9	3666.9
HCC Survival	XGBoost	0.8182	0.8346	0.0	18.7
HCC Survival	CatBoost	0.7879	0.8192	0.6	46.6
HCC Survival	xRFM	0.7576	0.8000	0.1	50.5
IDA2016	CatBoost	**0.9937**	0.9918	3.1	**0.5**
IDA2016	XGBoost	0.9929	**0.9928**	52.1	30.2
IDA2016	RF	0.9918	0.9893	5.8	15.1
IDA2016	xRFM	0.9897	0.9898	16.9	5.9
Crop Mapping	xRFM	**0.9945**	**0.9999**	12.8	**2.6**
Crop Mapping	XGBoost	0.9910	0.9999	40.0	19.6
Crop Mapping	RF	0.9892	0.9998	12.2	23.6

Seeds fixed at 42. TabPFN-v2 omitted throughout (see §2.3 for rationale). CatBoost on Crop Mapping timed out on our 2-hour Modal budget at 7-class × 30k × 174-feature scale and is omitted; the other

three models on Crop Mapping all converge. "Best" per dataset in bold; ties broken by AUC for binary, Accuracy for multiclass.

Observations on the numbers:

xRFM wins narrowly on the smooth-target datasets. On Appliances Energy (RMSE 65.29 vs XGBoost's 65.75) and Crop Mapping (Acc 0.9945 vs XGBoost's 0.9910) — both regimes where the target varies continuously with the inputs and AGOP re-weighting can pay off. On Crop Mapping the win is also Pareto-dominant: 2.6 μs/sample inference is $\approx$ 8–15$\times$ faster than XGBoost or RF.

The GBDTs win where rules and small-n favour them. CatBoost narrowly leads on Seoul Bike (all four models within 9% of each other) — the rule-like target (zero rentals when the system is off, season $\times$ hour interactions) plays to tree splitting. On HCC Survival, Random Forest leads xRFM by 9 accuracy points; with only 33 test samples, differences under $\pm$6 accuracy-points are not statistically distinguishable at the 5% level.

IDA2016 is the HP-sensitivity story. Tuned xRFM reaches AUC 0.990, within 0.3 AUC-points of XGBoost (0.9928) and CatBoost (0.9918). The scaling study in §3.3, which uses *default* HPs to isolate the effect of n, gives xRFM only 0.978. The 1.2-point tuned-vs-default gap is much larger than the 0.3-point xRFM-vs-GBDT gap, so most of xRFM's apparent loss is HP sensitivity, not algorithmic inferiority. We return to this in §4.

3.2 Interpretability on Appliances Energy

We use Appliances Energy as our interpretability testbed for two reasons. First, it includes two ground-truth random features (`rv1`, `rv2`) injected by the dataset's authors, which gives us a direct test of whether each importance method correctly ranks pure noise as low importance. Second, the 27 features have physical meanings (kitchen humidity, outdoor temperature, and so on), which lets us sanity-check the rankings against domain intuition.

Setup. We extract per-leaf AGOP diagonals from the xRFM trained with the Optuna-tuned hyperparameters from §3.1 (`kernel='l1', bandwidth=12.72, exponent=0.79, reg=0.061, diag=True`). With `diag=True` the Mahalanobis matrix is constrained to be diagonal, so we report only per-feature sensitivities — the paper's signed-eigenvector analysis (e.g., Fig. 7C on Breast Cancer) requires `diag=False` and is left as a follow-up. We compare AGOP against three alternatives chosen to span the importance-method design space: PCA loadings (an unsupervised baseline), mutual information (model-free, supervised), and signed permutation importance computed on an XGBoost proxy (black-box, supervised). Figure 1 ranks the top-15 features by AGOP and overlays the other three methods. Figure 2 shows AGOP per leaf,

replicating the multi-leaf comparison the paper performs on NYC Taxi data (Beaglehole et al. 2025, Fig. 7B).

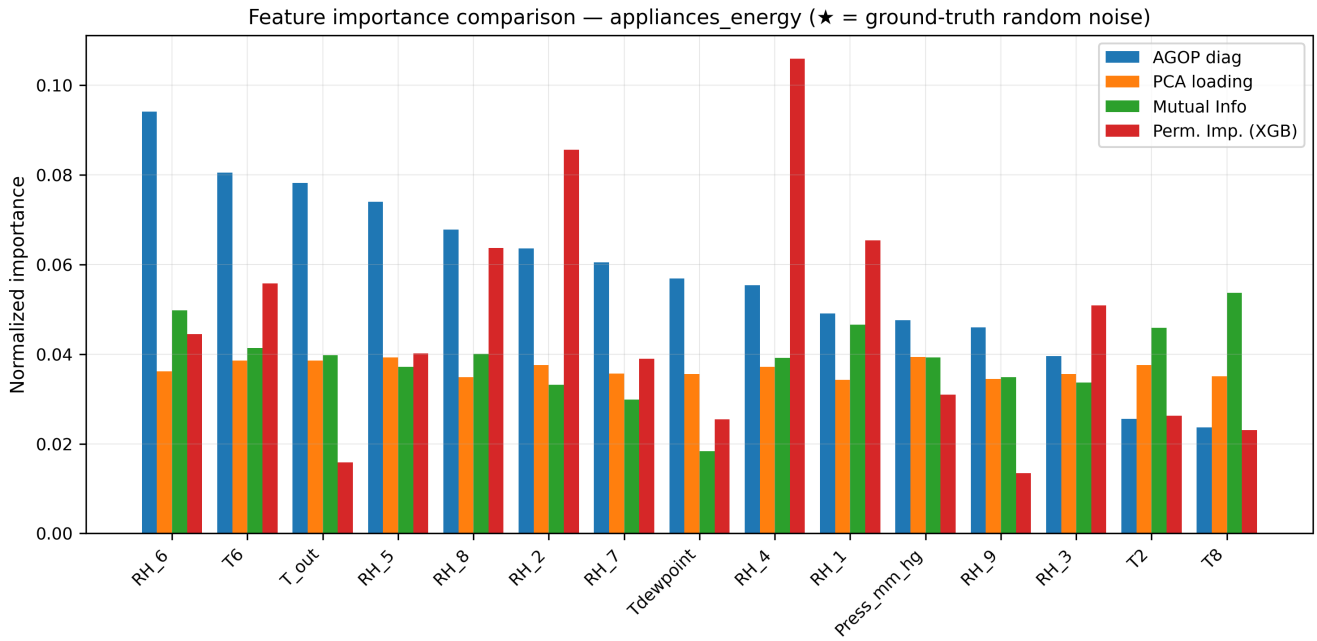

Figure 1: Feature importance comparison on Appliances Energy. [*] marks the two injected random features.

The random-feature test. The cleanest cross-method comparison is how each one ranks `rv1` and `rv2` — features that, by construction, carry zero signal. Out of $d = 27$ features:

Method	Rank of rv1	Rank of rv2
AGOP diagonal (tuned-HP xRFM)	21 / 27	20 / 27
PCA loading	**2 / 27** (wrongly top)	**1 / 27** (wrongly top)
Mutual information	26 / 27 (correct)	27 / 27 (correct)
Permutation (XGBoost, signed)	27 / 27 (correct)	26 / 27 (correct)

Mutual information and permutation cleanly identify the random features as least important. AGOP correctly puts them in the bottom third, but is less decisive than the model-free methods. PCA fails outright, ranking the noise features as the *most* important — because random uniforms are statistically independent of the other (correlated) sensor readings, they contribute disproportionately to the input covariance.

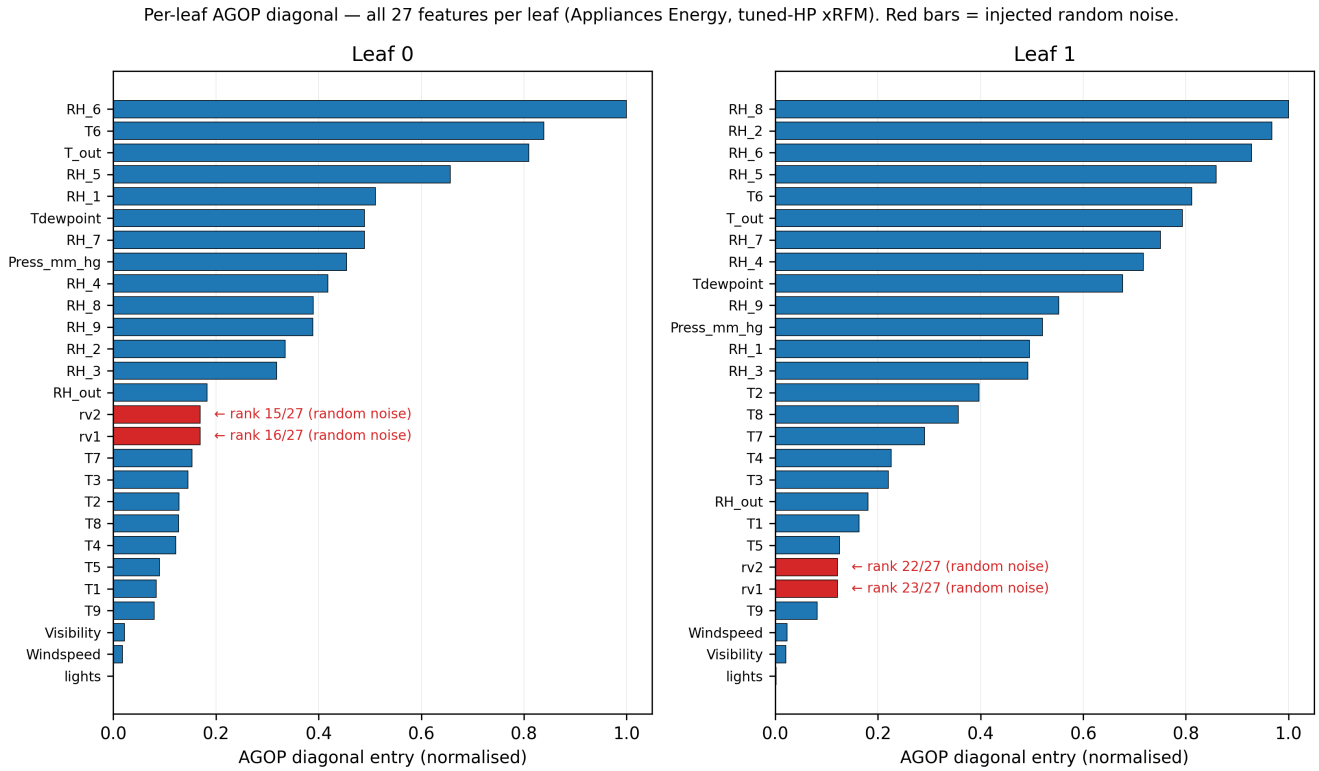

Figure 2: Per-leaf AGOP diagonal for the tuned-HP xRFM. The two leaves identify different humidity sensors as locally most important: RH_6 in leaf 0 and RH_8 in leaf 1.

(AGOP and PCA each assign ***rv1****,* ***rv2*** *exactly equal scores, so the rank distinctions in the table are tie-break artefacts. Per-leaf AGOP ranks are 15–16/27 in leaf 0 and 22–23/27 in leaf 1, so noise demotion strengthens with deeper splits.)*

What the random features tell us about AGOP. Two findings sit beneath the ranks. First, AGOP gives `rv1` and `rv2` exactly the same score because the model treats them as interchangeable noise — the AGOP iteration converges to the same diagonal entry by symmetry. The non-zero value (0.169 in leaf 0, 0.122 in leaf 1) is a residual that a finite-iteration kernel-ridge solver cannot drive to zero: the gap between "model-aware" and "model-free" measures (MI assigns clean zeros). Second, AGOP's noise rejection depends on the underlying kernel fit: re-running with default HPs (`bandwidth=10, kernel='l2', diag=False`) moves `rv1`, `rv2` to ranks 12–13 instead of 20–21 — a nine-position drop in discrimination. AGOP only sees what the model sees, so we use it to interpret what a *tuned* model has learned, not as a model-free importance measure.

Per-leaf structure shows subpopulation-specific features. The two leaves of the tuned tree concentrate on different humidity sensors: leaf 0 weights RH_6 (kitchen humidity) maximally (AGOP = 1.00), while leaf 1 weights RH_8 (parents' room humidity) maximally (= 1.00). T6 (outdoor temperature) and

RH_5 (bathroom humidity) are high in both. This mirrors the pattern Beaglehole et al. (2025, Fig. 7B) report for NYC Taxi, where different tree leaves discovered different locally-relevant features. With only two leaves on Appliances we see the effect on a small scale; deeper trees on larger datasets would amplify it.

Where the methods agree on real features. Both supervised methods that pass the noise test (AGOP and signed permutation) identify a humidity-dominated mix of sensors. AGOP's top features are RH_6, T6, T_out, RH_5, RH_8; permutation's top-5 are RH_4, RH_2, RH_1, RH_8, T4 — four of those five appear in AGOP's top-10. They disagree on which humidity sensor matters most but agree on the qualitative story: indoor humidity, not outdoor temperature alone, dominates the energy signal.

3.3 Scaling study on IDA2016

Figure 3 plots test AUC-ROC and training time against subsampled training size $n \in \{1k, 2.5k, 5k, 10k, 20k, 36k\}$, with the test set held fixed.

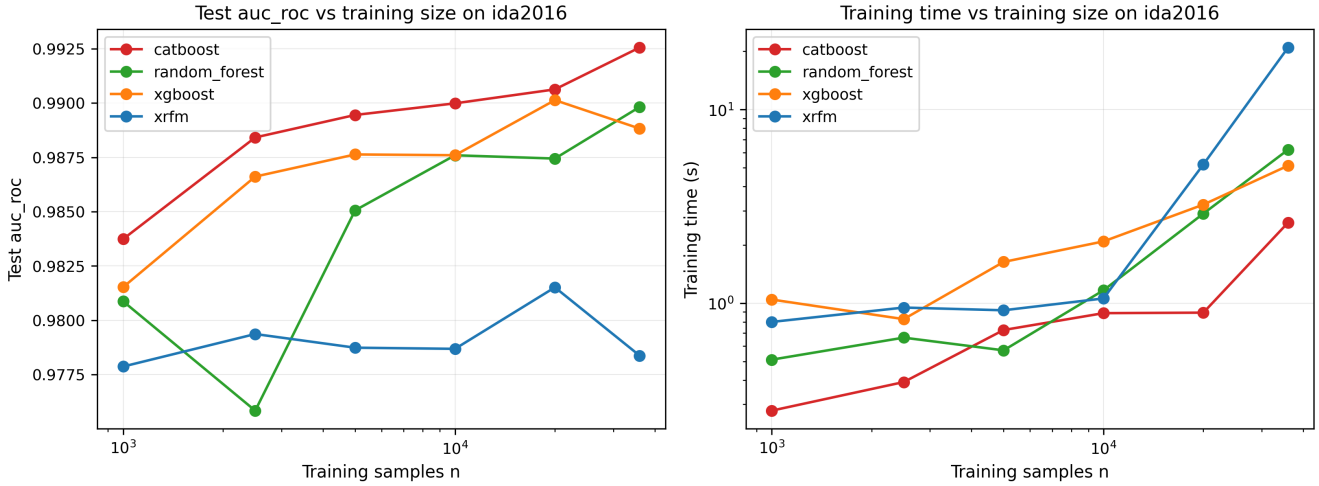

Figure 3: Test AUC-ROC (left) and training time (right) vs. training size on IDA2016.

Important: all four models use fixed (non-tuned) hyperparameters in this study, to isolate the effect of n. xRFM's corresponding *tuned* result is AUC 0.990 (Table 3), so the values below overstate the xRFM / GBDT gap relative to the main comparison. Three observations:

1. **Untuned asymptotic performance.** CatBoost reaches AUC 0.993, XGBoost and Random Forest $\approx$ 0.99, and **xRFM plateaus at 0.978** with default HPs. The gap narrows to 0.3 AUC-points under HP tuning (Table 3); most of the apparent loss is HP sensitivity, not algorithmic inferiority.

2. **Sample efficiency.** All four models are within 1.5 AUC-points of their asymptote at $n = 1{,}000$; this dataset's decision surface is fundamentally learnable from few samples. CatBoost and XGBoost

improve slightly with more data; default-HP xRFM shows essentially flat scaling behaviour.

3. **Training time.** xRFM training time climbs from 0.8 s at $n = 1k$ to 20.9 s at $n = 36k$, a 26.1× growth ratio over a 36× increase in n. The endpoint log-log slope is 0.91; the full-range fit is 0.84. The paper's $O(n \log n)$ claim would predict local slopes of 1.10–1.15 and a growth ratio near 54× over the same range, so our measured timings are *sub-linear* (closer to $O(n^{0.84})$) — **not** $O(n \log n)$. We attribute this to operating in the single-leaf regime: at $L_{\max} = 60{,}000 \geq n_{\text{train}}$ throughout this study, xRFM never partitions and the inner kernel-ridge solve dominates training. The asymptotic $O(n \log n)$ claim is about the regime where the tree splits, which our experiment does not enter.

4. Discussion

xRFM's results match theory on four datasets, with one surprise. The theoretical argument for feature-learning kernel methods is that they should win when the target varies *smoothly* with the inputs — small changes in features producing small changes in the output — because the model can profit from re-weighting the directions that matter. We see this on Appliances Energy, where energy consumption varies continuously with humidity and temperature readings: xRFM beats every GBDT by a hair. The same story holds on Crop Mapping, where the 174 satellite bands are physically continuous and correlated, and xRFM's AGOP re-weighting again edges out the trees. The mirror image is Seoul Bike: rentals follow sharp rules (zero when the system is off) and multiplicative interactions (season × hour), exactly the structure trees split on naturally — and the GBDTs win, narrowly.

The surprise is HCC Survival. Classical kernel theory says kernels should dominate trees in the small-n regime, since RKHS regularisation provides a strong prior. We expected xRFM to win here. Instead Random Forest beats it by 9 accuracy points. We suspect the cause is AGOP estimation: with only 99 training samples, learning a 49×49 feature-importance matrix is statistically fragile, and a noisy AGOP feeds back into a noisy reweighting that hurts more than it helps.

xRFM is more hyperparameter-sensitive than the GBDTs. IDA2016 makes this clearest. With HP tuning, xRFM achieves AUC 0.990 — within 0.3 AUC-points of CatBoost and XGBoost. The scaling study in §3.3 (which used *default* HPs to isolate the effect of n) gives xRFM AUC 0.978, a 1.2-point drop. CatBoost's defaults sit within 0.1 AUC-points of its tuned result on the same data, so the asymmetry is real. Three structural features of IDA2016 stress kernel methods more than trees, and each matches one of Grinsztajn et al. (2022)'s predictions about why kernels lose on tabular data:

- **Severe class imbalance (1.6% positive).** Kernel ridge regression minimises squared error symmetrically, so it has no built-in mechanism to up-weight the rare class. GBDTs can be told to

up-weight the minority directly. xRFM can compensate through bandwidth and ridge tuning, but only if those HPs are actually searched.

- **170 dimensions, most low-signal.** xRFM has to *learn* via AGOP which features matter; trees ignore irrelevant features for free through split selection.
- **Heavy missingness (8.3% of cells).** Trees absorb missingness through their split logic — a missing value is just another route through the tree. A smooth kernel needs imputation to be 'right'; wrong imputation directly distorts the similarity measure between points.

AGOP's strength and its cost. AGOP inherits the model's learned feature sensitivity, which cuts both ways: a well-tuned RFM produces a discriminating AGOP that correctly demotes random noise to ranks 20–21 of 27; a badly-tuned RFM produces an AGOP that places the same noise at 12–13. Mutual information and signed permutation sidestep this dependency. The takeaway: **AGOP for interpreting a tuned model, MI or signed permutation as a model-free sanity check, PCA as a null baseline.**

Limitations and next steps.

- **Single-seed results.** All experiments use one fixed seed (42). Small differences — especially on HCC's 33-sample test set — may not survive multi-seed replication. Multi-seed runs would let us put error bars on every comparison.
- **Restricted kernel-family search.** The paper's full kernel family is $K_{p,q}$ for $0 < q \leq p \leq 2$, a 2D grid over (p, q). Our HP search varies only a single `exponent` parameter and restricts kernel choice to two slices: `l1` ($K_{p,p}$, axis-aligned) and `l2` ($K_{p,2}$, rotation-invariant). The tuned HPs picked `l1` on Appliances Energy and Seoul Bike, `l2` on the other three datasets (`results/downloads/main_*_xrfm_seed42.json`), so both slices saw use — but sweeping the full (p, q) grid is the natural follow-up.
- **Single-leaf regime throughout.** Our `max_leaf_size = 60{,}000` is at least as large as n_{train} on every dataset, so xRFM never actually partitions; it operates as a single-leaf RFM with AGOP feature learning. This means the tree component of "xRFM" is never engaged in our experiments. Our results therefore characterise the leaf RFM, not the full algorithm — and the paper's $O(n \log n)$ training claim is about the partitioning regime our experiments do not enter (consistent with the sub-linear scaling we observed in §3.3).
- **TabPFN-v2 omitted.** TabPFN-v2 would be a useful third baseline (a tabular foundation model, neither tree nor kernel). Its license gates weight download behind interactive acceptance, which blocked our headless Modal pipeline.

- **Shallow interpretability tree.** Both our default-HP and tuned-HP interpretability runs produced only 2 leaves. The richer per-leaf comparison the paper performs on NYC Taxi (Beaglehole et al. 2025, Fig. 7A, with 8 leaves) would require a larger dataset to induce deeper trees.

5. Conclusion

xRFM fills a real gap in the tabular kernel-methods literature as the first feature-learning kernel machine to scale past roughly 70k samples. On our five UCI datasets it matches or exceeds GBDTs on smooth-target tasks (Appliances Energy, Crop Mapping), is competitive with HP tuning on imbalanced high-dimensional classification (IDA2016), and trails on small-n (HCC) and rule-like targets (Seoul Bike) where tree inductive bias is better matched. Its practical differentiator is per-leaf AGOP interpretability — a *supervised* alternative to PCA and a *model-aware* alternative to mutual information — though our random-feature test shows AGOP's noise rejection depends heavily on the quality of the underlying kernel fit. xRFM is a credible complement to GBDTs, especially when interpretability matters, but has not dethroned XGBoost or CatBoost as a reliable first choice.

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Appendix A: Dataset Feature Descriptions

A.1 Seoul Bike Sharing (UCI 560)

Hourly rental counts in Seoul, Sept 2017 – Nov 2018 ($n = 8{,}760$, $d = 12$). Target: `Rented_Bike_Count` $\in [0, 3556]$, mean 704.6, std 645.0. **No missing values.**

- **Numeric (9):** `Hour` (0–23), `TemperatureC` (−17.8 to 39.4), `Humiditypct` (0–98), `Wind_speed_m/s` (0–7.4), `Visibility_10m` (27–2000), `Dew_point_temperatureC` (−30.6 to 27.2), `Solar_Radiation_MJ/m²` (0–3.52), `Rainfallmm` (0–35), `Snowfall_cm` (0–8.8).
- **Categorical (3):** `Seasons` (4 levels), `Holiday` (Yes/No), `Functioning_Day` (Yes/No).

The target follows clear rules: rentals are exactly zero whenever the bike system is off (`Functioning_Day=No`); otherwise demand has two daily peaks (morning and evening commutes); and the time-of-day pattern depends strongly on the season. These are exactly the kind of axis-aligned, piecewise-constant patterns that decision trees handle naturally — a structural reason to expect GBDTs to do well here.

A.2 Appliances Energy Prediction (UCI 374)

10-minute energy readings from a Belgian house ($n = 19{,}735$, $d = 27$, all numeric). Target: `Appliances` (Wh) $\in [10, 1080]$, mean 97.7, std 102.5. **No missing values.**

- **Indoor sensors (18):** `T1`–`T9` (room temperatures, 14.9–29.9 °C), `RH_1`–`RH_9` (room humidities, 1–100%).
- **Outdoor weather (7):** `T_out`, `Press_mm_hg`, `RH_out`, `Windspeed`, `Visibility`, `Tdewpoint`, `lights`.
- **Random noise (2):** `rv1`, `rv2`, both drawn from $\mathcal{U}(0, 50)$ by Candanedo et al. (2017) as ground-truth-irrelevant features. Used as the "noise rejection" check in §3.2.

A.3 HCC Survival (UCI 423)

Hepatocellular carcinoma 1-year survival in a Portuguese cohort ($n = 165$, $d = 49$). Target: binary survival outcome, **class balance 102 lives : 63 dies (62%/38%). 826 missing values (10.2% of cells).** Features anonymised as `x0`–`x48`.

- **Categorical (27):** `x0`–`x22` are 22 binary clinical flags (cirrhosis, alcoholism, hepatitis, ascites, etc.); `x26` (5 levels: ECOG performance status), `x27`/`x28` (3 levels each: encephalopathy/ascites grades), `x43` (6 levels: Child-Pugh).
- **Numeric (22):** Lab values with widely varying scales: `x23` (age, 20–93), `x30` (likely AFP, range 1.2 to 1.8M), `x33`/`x34` (likely AST/ALT, max 13k and 459k respectively), plus haemoglobin, platelets, bilirubin, INR, creatinine, iron studies. Missingness up to 48% on rarely-ordered tests.

High $d/n \approx 0.30$, which is the report's primary hypothesis for xRFM's underperformance on this dataset.

A.4 IDA2016 Scania APS (UCI 414)

Heavy-truck Air Pressure System failure prediction ($n = 60{,}000$, $d = 170$, all numeric). Target: binary (failure/no), **class balance 59,000 : 1,000 (1.67% positive). 850,015 missing values (8.3% of cells).**

Features are anonymised sensor readings (`aa_000`, `ab_000`, …); the `ag_000`–`ag_009` series are histogram bins of one underlying counter. **Eight columns are >50% missing** (e.g., `br_000` 82%, `bq_000` 81%, `bp_000` 80%). All three properties stress kernel methods specifically. With 170 mostly-uninformative dimensions, xRFM has to *learn* which features matter, whereas trees ignore irrelevant features for free via split selection. With only 1.6% positive class, kernel ridge regression has no built-in mechanism for asymmetric error costs, where GBDTs can be told to up-weight the minority. And with 8.3% of cells missing, tree-based methods absorb missingness through their split logic, whereas a smooth kernel needs the imputation to be 'right'. We expected this dataset to be the hardest of our five for xRFM, and it was.

A.5 Crop Mapping, Winnipeg (UCI 525)

Crop classification from fused satellite data over Manitoba, 2015–2016 ($n = 325{,}834$, stratified-subsampled to $n = 50{,}000$; $d = 174$, all numeric, no missing). Target: 7 crop classes, with **highly skewed prevalence** (largest class 26.1%; smallest 0.35%, only 175 samples).

Features come from two satellite systems collected at multiple dates: **RADARSAT-2 synthetic aperture radar** (4 polarisation channels — HH, HV, VH, VV — plus derived parameters) and **Landsat-8 optical bands** (visible, near-infrared, short-wave infrared). Raw values span roughly $[-28{,}400, 1{,}089{,}300]$

— the radar measurements are in negative-dB units while optical radiances are large positive numbers — so standardisation is essential. Crucially for our hypothesis, all 174 features are continuous and physically correlated: the kind of smooth, structured feature space where a kernel's similarity-based assumption matches the data well.